

■ Mobility Lens

CS501 E1 — Mobile Application Development — Assignment 1

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Package	com.priyanshu.mobilitylens	Emulator	Pixel XL — API 33

■■ SDK Settings

minSdk is 24. I picked 24 so I was not stuck supporting really old phones, and Compose still works fine from that version.

targetSdk is 35 because that is the current release, and it is why I called `enableEdgeToEdge()` in MainActivity — that behavior is tied to 35.

compileSdk is also 35. That is just what the compiler builds against. The Pixel XL I tested on is API 33 and it ran the app fine — minSdk is what controls that, not compileSdk.

■ Key Project Files

MainActivity.kt

app/src/main/java/com/priyanshu/mobilitylens/

This is the screen. All the composables — the dimension card, the Previous and Next buttons, the text field, the Analyze button — are defined here. State lives here too, with remember. MobilityDimension.kt next to it just holds the three resource IDs per dimension.

AndroidManifest.xml

app/src/main/

Launcher activity, the theme, that is basically it. No extra permissions because the app never touches the network, camera, or location. I did not add INTERNET either — it has to be explicitly declared and I did not need it.

build.gradle.kts

app/

SDK numbers, Compose flag, and the library dependencies are all here. References libs.versions.toml for the actual version numbers.

libs.versions.toml

gradle/

I put all versions here so I am not typing 1.15.0 in five different places. build.gradle.kts just references the aliases.

strings.xml

app/src/main/res/values/

The six dimension names, descriptions, implications, and all the button labels are here instead of hardcoded in Kotlin. Made it easier to edit copy without touching the layout code.

■ Rotation Observation

I was on Dimension 3, typed Snapchat in the text field, then rotated the emulator to landscape. The app landed back on Dimension 1 and the field was empty. I used `remember` for `currentIndex` and `appNameInput`. `remember` only holds state while the composable is alive — rotation kills and recreates the Activity, so `Compose` starts fresh and both values go back to their defaults.

I did not fix it because the assignment said not to. But it matters on a real phone. Say someone is on Dimension 4, types their app name, then their phone rotates because they tilted it to watch a video or took a call and came back. Everything they did is gone. `rememberSaveable` would write those values into the saved instance state and restore them after the Activity comes back.

■ Mobile Design Decision

I gave each dimension its own accent color — the progress bar, the emoji circle, and the Next button all update together when you navigate. I did this because the screen is small and there is not much room for extra labels telling the user something changed. The color shift does that instantly. Even if someone cannot distinguish the colors, the emoji and the 'Dimension 2' label still update, so it is not color-only feedback. One-handed use was also in my head — the Next button is big and in the bottom half of the screen where a thumb can reach it.

■ Collaboration and Generative-AI Disclosure

Nobody to list. I worked on this alone.

I used Claude for setup help, fixing build errors, and getting the project structure going. I did not let it write the six dimension descriptions — I rewrote those myself and I can explain any file if asked.